Summary, Outlook, and the Proofs behind the Riccati Tests

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The last chapter of the notes is one page of summary and three short appendices, but the appendices carry the weight: they contain the proof of the equivalence between "the Hamiltonian has no imaginary eigenvalues", "the Riccati equation has a stabilizing solution" and "the Riccati inequality is solvable", which every synthesis formula in chapters 7 and 8 rents without owning. We gather the summary into a map of what the notes built and what they left open, write out the bisection algorithm of Appendix A and run it on the scalar example of chapter 7 (it returns $\sqrt{3}-1$ to eight digits), prove the spectral symmetry of a Hamiltonian matrix, and then go through the proof of Theorem 7.2 in full: the difference identity, the minimality of the stabilizing solution, the Lagrangian property of the stable invariant subspace that makes it a graph, and the perturbation argument that turns a Riccati inequality into a Riccati equation. Theorem 7.1 follows by the sign trick of Appendix B. The chapter ends with the outlook the notes themselves give: what LMI techniques add beyond the H_∞ bounds used here.

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1. What the notes built, and what they left open

The book's own summary is four paragraphs, and it is worth restating as a map, because the map is more useful than any individual result.

> 1. **The framework is not the contribution.** The generalized plant captures almost any interconnection, and it extends — with less generality — beyond linear systems. Everything in the notes is a statement about one plant P and one closed loop $P \star K$.
> 2. **Analysis is solved for LTI uncertainties.** The structured singular value gives computable robust stability and robust performance tests against structured LTI uncertainties. Parametric (real) uncertainties can be included, but the computation becomes delicate — the exact μ problem is NP-hard, so only bounds are computed.
> 3. **Nominal synthesis is solved.** Minimizing an H_∞ performance channel has a complete solution in terms of two Riccati equations and a coupling condition (chapter 7), and every sub-optimal controller is $K_{\text{par}} \star Q$ (chapters 7 and 9).
> 4. **Robust synthesis is not solved.** Minimizing the structured singular value over controllers has no satisfactory algorithm; the scalings/controller iteration (chapter 8) is the only general method, it gives no optimality guarantee, and it works well in practice often enough to be worth using.
> 5. **The bounds are the limitation.** The notes use H_∞ norm bounds as the whole performance language; the outlook (section 6) is what changes when that restriction is lifted.

2. Appendix A: bisection, and how a yes/no test becomes a number

Every "compute γ_{critical}" in these notes is really a **decision** problem: for a given γ, either the test passes or it fails. Computing an H_∞ norm (Theorem 7.4), computing the optimal H_∞ level (Theorem 7.13) and computing the μ upper bound (chapter 4) are all of this shape. One algorithm converts a decision into a number, and it is worth writing out because it is used everywhere.

> **Bisection (Appendix A).** Fix $\varepsilon > 0$ and an initial interval $[a_1, b_1]$ with $a_1 \le \gamma_{\text{critical}} \le b_1$. Given $[a_j, b_j]$, test $\gamma_{\text{critical}} < (a_j + b_j)/2$. If the test says yes, take $[a_{j+1}, b_{j+1}] = [a_j, (a_j + b_j)/2]$; if it says no, take $[a_{j+1}, b_{j+1}] = [(a_j + b_j)/2, b_j]$. Stop when $b_{j+1} - a_{j+1} \le \varepsilon$, and report any point of the interval.

Each step halves the interval, so the number of steps grows like $\log_2((b_1 - a_1)/\varepsilon)$: doubling the accuracy costs one extra test. The only requirement on the underlying decision problem is that it be **monotone** — that pass/fail switch exactly once as γ grows. That is what the theorems supply: Theorem 7.13 fails below the optimal level and holds above it, so bisection applies.

As a check we run it on the scalar plant of chapter 7, whose optimal level we know in closed form: the two indefinite Riccati equations collapse to one quadratic, the coupling condition becomes $X_\infty < \gamma$, and solving $X_\infty = \gamma$ gives $\gamma_* = \sqrt{3} - 1$. Bisecting on the decision "both Riccati equations have stabilizing solutions with $\rho(X_\infty Y_\infty) < \gamma^2$" from $[0.6, 5]$ gives

j	a_j	b_j	$b_j - a_j$
1	0.600000000	5.000000000	4.4e0
5	0.600000000	0.875000000	2.8e-1
10	0.728906250	0.737500000	8.6e-3
15	0.731860352	0.732128906	2.7e-4
20	0.732044983	0.732053375	8.4e-6
22	0.732049179	0.732051277	2.1e-6

,

with final midpoint 0.732050753 against the exact $\sqrt{3} - 1 = 0.732050808$ — an error of $5.5 \cdot 10^{-8}$ after 22 tests. The script is `scripts/bisection.m`. The example also shows what bisection does **not** do: it returns a number, not a controller, and it cannot tell which of the three conditions of Theorem 7.13 failed for a γ below the optimum.

3. The shape of a Hamiltonian matrix

The proofs in Appendices B and C use one structural fact, and it explains why the Riccati theory is a theory about the imaginary axis.

1. The **Hamiltonian** structure is JH symmetric, where
$$J = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}. \tag{1}$$
2. *Consequence* (**Lemma C.1**): if H has k eigenvalues in $\mathbb{C}_<$ it has exactly k in $\mathbb{C}_>$ (with multiplicity), and vice versa.

The proof of the lemma is three lines and is the kind of computation worth doing once. With J as above, $JH = (JH)^T = H^T J^T = -H^T J$, so $JHJ^{-1} = -H^T$: the matrix H is **similar** to $-H^T$. Similar matrices have the same characteristic polynomial, so

$$\det(H - sI) = \det(-H^T - sI) = \det(H^T + sI) = \det(H + sI) = \det(H - (-s)I), \tag{2}$$

using that H has even size and that $\det(-M) = (-1)^{2n}\det(M) = \det(M)$. So the spectrum of H is symmetric about the origin of the imaginary axis: if λ is an eigenvalue of multiplicity k, so is $-\lambda$. This is why a Hamiltonian matrix either has eigenvalues on the imaginary axis or splits them evenly into n stable and n unstable ones — and only in the second case can the Riccati equation be solved.

4. Appendix C: the proof of Theorem 7.2

We prove: for symmetric Q, $R \geq 0$ and stabilizable (A, R), the following are equivalent — (a) the Hamiltonian $H = \begin{pmatrix} A & R \\ -Q & -A^T \end{pmatrix}$ has no eigenvalue on the imaginary axis; (b) the ARE $A^T X + XA + XRX + Q = 0$ has a (unique) stabilizing solution X_-; (c) the ARI $A^T X + XA + XRX + Q < 0$ has a symmetric solution. If so, every solution of the ARE or ARI satisfies $X_- \leq X$.

4.0.1. Step 0: the difference identity, and minimality of X_-

Identity (C.1). For any two symmetric Z, Y,

$$\begin{aligned} &(A^T Z + ZA + ZRZ + Q) - (A^T Y + YA + YRY + Q) \\ &= (A + RY)^T (Z - Y) + (Z - Y)(A + RY) + (Z - Y)R(Z - Y). \end{aligned} \tag{3}$$

This is Lemma 7.24 of chapter 7 and it is verified by expanding the right-hand side. Its importance is that it is **linear** in the difference $Z - Y$ except for one term, which is ≥ 0 when $R \geq 0$.

Take $Z = X$ (any solution of the ARE or ARI, so the left side is ≤ 0) and $Y = X_-$ (a stabilizing solution, so the left side is exactly 0 if X solves the ARE and ≤ 0 if it solves the ARI). Dropping the non-negative term $\Delta R \Delta$ leaves

$$0 \geq (A + RX_-)^T \Delta + \Delta (A + RX_-), \quad \Delta := X - X_-. \tag{4}$$

Since $A + RX_-$ is stable, $\Delta \geq 0$: the stabilizing solution is the **smallest** one. Two consequences follow immediately. First, X_- is the unique stabilizing solution: if X_1, X_2 both stabilize then each is the smallest, so $X_1 \leq X_2$ and $X_2 \leq X_1$. Second, the ARI cannot be solvable while the ARE has no stabilizing solution below it, which is half of (c) $\to$ (b).

4.0.2. Step 1: (a) implies (b)

This is the constructive direction and the one worth reading closely. Since H has no imaginary eigenvalues, Lemma C.1 splits its spectrum evenly: n eigenvalues in $\mathbb{C}_<$ and n in $\mathbb{C}_>$. Take an ordered Schur decomposition

$$T^* H T = \begin{pmatrix} M & M_{12} \\ 0 & M_{22} \end{pmatrix} \tag{5}$$

with M $(n \times n)$ stable and M_{22} anti-stable, and let Z be the first n columns of T, so that

$$HZ = ZM. \tag{6}$$

Partition $Z = \begin{pmatrix} U \\ V \end{pmatrix}$ into two $n \times n$ blocks. The technical heart is to show that U is invertible; then $X_- := VU^{-1}$ is the stabilizing solution. Three facts are proved in order.

(i) The stable subspace is Lagrangian: $V^*U = U^*V$. From $HZ = ZM$ we get $Z^* J H Z = Z^* J Z M$. The left side is Hermitian (because JH is symmetric), so the right side is too: taking adjoints and using $J^* = -J$ and $(Z^* J Z)^* = -Z^* J Z$,

$$(Z^*JZ)M = (Z^*JHZ)^* = M^*(Z^*JZ)^* = -M^*(Z^*JZ). \tag{7}$$

Rearranged this is the Lyapunov equation $M^*(Z^*JZ) + (Z^*JZ)M = 0$, whose only solution is zero because M is stable. Expanding $Z^*JZ = 0$ with $J = \left(\begin{smallmatrix} 0 & -I \\ I & 0 \end{smallmatrix}\right)$ gives exactly $V^*U = U^*V$.

(ii) $Ux = 0$ implies $RVx = 0$ and $UMx = 0$. If $Ux = 0$, the first block row of $HZ = ZM$ reads $UMx = (AU + RV)x = RVx$. Left-multiplying by x^*V^* and using (i) plus $Ux = 0$ turns the left side into x^*V^*RVx and the other route into 0; since $R \geq 0$ that forces $RVx = 0$, and then $UMx = 0$ as well.

(iii) U is invertible. If not, (ii) shows $\ker(U)$ is a nonzero M-invariant subspace, so it contains an eigenvector: there is $x \neq 0$ with $Mx = \lambda x$ and $Ux = 0$. The second block row of $HZ = ZM$ reads $(-QU - A^TV)x = VMx = \lambda Vx$, i.e. $A^TVx = -\lambda Vx$. Since M is stable, $\Re(-\lambda) > 0$, and (ii) gave $R^TVx = 0$; stabilizability of (A, R) now forces $Vx = 0$. With $Ux = 0$ this makes $Zx = 0$, hence $x = 0$ because Z has full column rank — a contradiction.

So U is invertible. Because $V^*U = U^*V$, the matrix $X_- = VU^{-1}$ is Hermitian: $\left(VU^{-1}\right)^* = U^{-*}V^* = VU^{-1}$ using (i). To see that X_- solves the ARE and stabilizes, divide $HZ = ZM$ by U on the right:

$$H\begin{pmatrix} I \\ X_- \end{pmatrix} = \begin{pmatrix} I \\ X_- \end{pmatrix}(UMU^{-1}). \tag{8}$$

The first block row gives $A + RX_- = UMU^{-1}$, which is stable. The second block row gives $-Q - A^TX_- = X_-(A + RX_-)$, which is the ARE. Finally, X_- is **real**: the data are real, so conjugating the ARE and the stability condition shows $|X|_-$ is also a stabilizing solution, and uniqueness forces $X_- = |X|_-$.

4.0.3. Step 2: (b) implies (a)

A one-line similarity. If X_- solves the ARE and $A + RX_-$ is stable,

$$H\begin{pmatrix} I & 0 \\ X_- & I \end{pmatrix} = \begin{pmatrix} I & 0 \\ X_- & I \end{pmatrix}\begin{pmatrix} A + RX_- & R \\ 0 & -(A + RX_-)^T \end{pmatrix}. \tag{9}$$

So H is similar to a block-triangular matrix whose diagonal blocks are $A + RX_-$ and $-(A + RX_-)^T$, both stable or anti-stable: H has no imaginary eigenvalues. (This is also the identity behind the graph picture of chapter 7: the graph $\left(\begin{smallmatrix} I \\ X_- \end{smallmatrix}\right)$ is invariant under H.)

4.0.4. Step 3: (a) implies (c)

A continuity argument. Perturb $Q \to Q + \varepsilon I$ and call the Hamiltonian H_ε. Eigenvalues depend continuously on the entries of a matrix, so for all small $\varepsilon > 0$ the matrix H_ε still has no imaginary eigenvalues. By Step 1 applied to $Q + \varepsilon I$, there is a symmetric X with $A^TX + XA + XRX + Q + \varepsilon I = 0$, which gives the strict ARI (c).

4.0.5. Step 4: (c) implies (b)

Let Y satisfy the strict ARI and set $P \coloneqq A^T Y + YA + YRY + Q < 0$. The claim is that the ARE

$$(A + RY)^T \Delta + \Delta(A + RY) + \Delta R \Delta + P = 0 \tag{10}$$

has a **stabilizing** solution Δ. If it does, the difference identity (C.1) shows that $X \coloneqq Y + \Delta$ solves the original ARE and that $A + RX = (A + RY) + R\Delta$ is stable, which is (b). The existence of Δ is where chapter 7′s Lemma 7.3 is used: $(A + RY, R)$ is stabilizable, $R \geq 0$, and $P < 0$ makes (A, P) observable (since P is definite), so $(A + RY, P)$ has no unobservable mode on the imaginary axis; Lemma 7.3 then says the Hamiltonian of the Δ-equation has no imaginary eigenvalues, and Step 1 (already proved) supplies Δ. This closes the cycle (a) $\to$ (b) $\to$ (a) and (a) $\to$ (c) $\to$ (b), and the sandwich $X_- \leq X$ from Step 0 finishes the theorem. $\square$

5. Appendix B: Theorem 7.1 from Theorem 7.2

Theorem 7.1 differs from Theorem 7.2 only in assuming (A, R) **controllable** instead of stabilizable, and concluding the existence of the **anti-stabilizing** solution X_+ in addition to X_-. The reduction is a sign change: observe that

$$\begin{aligned} A^T X + XA + XRX + Q = (-A)^T(-X) + (-X)(-A) + (-X)R(-X) + Q, \\ A + RX = -((-A) + R(-X)). \end{aligned} \tag{11}$$

So "X is a stabilizing solution for (A, R, Q)" is the same statement as "$-X$ is an anti-stabilizing solution for $(-A, R, Q)$". Applying Theorem 7.2 with $-A$ in place of A gives:

> **Corollary B.1 (book).** Suppose the hypotheses of Theorem 7.2 hold but $(-A, R)$ is only stabilizable. Then (a) H has no imaginary eigenvalues, (b) the ARE has an anti-stabilizing solution X_+, and (c) the ARI has a symmetric solution are equivalent, and every solution satisfies $X \leq X_+$.

Controllability of (A, R) gives stabilizability of both (A, R) and $(-A, R)$, so Theorem 7.2 applied to (A, R) gives X_- with $X_- \leq X$, and Corollary B.1 gives X_+ with $X \leq X_+$: together, Theorem 7.1. The book notes that with only stabilizability of (A, R) the anti-stabilizing solution generally fails to exist — a subtlety the Riccati machinery of chapter 7 has to respect when it fixes which solution of each Riccati equation defines the central controller.

6. The outlook: what linear matrix inequalities add

The last page of the book is a list, and it is honest about the boundary of what was proved here: every result in these notes is about **LTI controllers, LTI uncertainties, and H_∞ bounds**. Relaxing each of the three is what the LMI framework is for.

1. **A richer performance language.** Instead of one H_∞ bound, one can impose H_2 constraints (stochastic noise reduction), positive-real (passivity) conditions, and amplitude constraints on selected channels, all in one optimization. The reason is that the closed-loop transfer matrix depends **affinely** on the controller once it is parametrized by the Youla parameter of chapter 9 — exactly the affine dependence $T_1 + T_2 Q T_3$ — so these requirements become convex, LMI-representable conditions in Q.
2. **Multi-objective design.** Because the constraints are convex in Q, different specifications on different channels can be traded off in a single problem instead of by iterating on weights.
3. **Uncertainties beyond LTI.** Time-varying parametric, nonlinear static and nonlinear dynamic uncertainties can be covered by **integral quadratic constraints**, which generalize the scalings of chapters 4 and 8: the scaling multiplier is replaced by a multiplier that certifies an integral inequality between the uncertainty's input and output. The D-scaling step of the scalings/controller iteration becomes a search over multipliers.
4. **Gain scheduling.** The same machinery gives a systematic way to design controllers that switch with a measured scheduling variable, with stability guaranteed over the schedule — the practical counterpart of the switching application listed in chapter 9.

> **Where the open problem is.** The single unresolved item in the notes is robust **synthesis**: minimizing the structured singular value over controllers. Everything else in the list above is a solved (convex) problem once the robustness requirement is dropped or replaced by a multiplier test. The scalings/controller iteration remains the only general method, and it is a coordinate descent, not an algorithm with a guarantee.

7. Sources and further reading

The only source read for this note is [1]: page 241 for the summary and outlook, Appendix A (page 242) for the bisection algorithm, Appendix B (pages 242–243) for the reduction of Theorem 7.1 to Theorem 7.2, and Appendix C (pages 243–246) for the proof of Theorem 7.2, including Lemma C.1 and the difference identity (C.1). The bisection table was produced by the note's own script `scripts/bisection.m` in Octave, not taken from the book.

Further reading, not consulted. [2] is the standard reference for the LMI framework and the multi-objective and IQC extensions described in the outlook section; [3] is the standard textbook for the Riccati theory proved in the appendices. Neither was read while writing this note.

Bibliography

[1] C. W. Scherer, "Theory of Robust Control." [Online]. Available: https://www.imng.uni-stuttgart.de/mst/files/RC.pdf°

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[3] K. Zhou, J. C. Doyle, and K. Glover, *Robust and Optimal Control*. Prentice Hall, 1996.